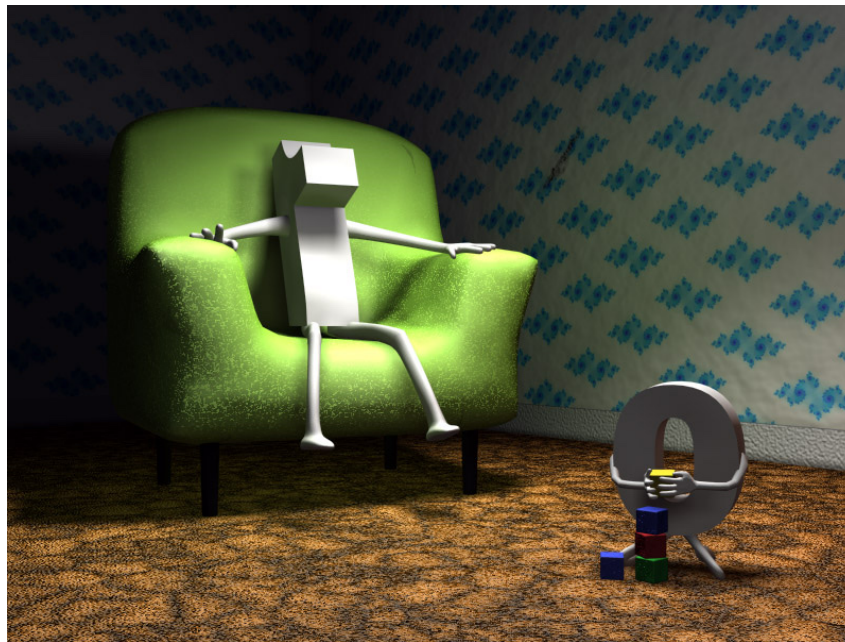


Chapter 1. INTEGERS

1. 2 Divisibility and Prime Factorisation

1.2.2 Gcd. Primes and coprime integers



Today we will talk

- Divisibility
- Common divisors
- Greatest common divisor (gcd)
- Coprimes (or relatively prime) integers

Let us revisiting a very important concept: divisibility. If n and d are two integers with $d \neq 0$. We say that n is **divisible by** d , or that d is a **divisor of** n , if there exists an integer q such that $n = qd$. We will write $d|n$. We also say that n is a **multiple of** d . If d does not divide n , we write $d \nmid n$.

Example 1.9. Observe that $2|108$, or 108 is divisible by 2 since $108 = 2 \cdot 54$. But $5 \nmid 108$ since there is no integer q for which $108 = 5q$.

★ **NOTE:** $1|n$ and $n|0$, for all integers n .
 $r = 0 \Leftrightarrow d|n$

THEOREM 1.10. Let m, n and d be integers.

- (1) $n|n$ for all n .
- (2) $d|m, m|n \Rightarrow d|n$.
- (3) $d|n, n|d \Rightarrow d = \pm n$.
- (4) $d|n, d|m \Rightarrow d|(xn + ym)$, for all integers x and y .

Proof of (4)

Suppose that $d|n$ and $d|m$. Then $n = q_1 d$ and $m = q_2 d$ for some integers q_1 and q_2 .

For any integers x and y we have

$$xn = (q_1 x) d \text{ and } ym = (q_2 y) d$$

$$xn + ym = (q_1 x + q_2 y) d \Rightarrow d | (xn + ym) \quad \blacksquare$$

Definition 1.12. Let m and n be integers. We say that an integer d is a **common divisor of m and n** if $d|m$ and $d|n$.

Example 1.13. Consider the positive divisors of 36 and 84:

POSITIVE DIVISORS OF 36

1, 2, 3, 4, 6, 9, 12, 18, 36

POSITIVE DIVISORS OF 84

1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84

COMMON DIVISORS OF 36 AND 84

1, 2, 3, 4, 6, 12

★ **NOTE!** The **largest common divisor 12** is a **multiple** of all the other positive common divisors.

Definition 1.14. Let m and n be integers, not both zero. An integer d is called the **greatest common divisor of m and n** , written $d = \gcd(m, n)$ if it satisfies the following:

- (1) $d \geq 1$.
- (2) $d|m$ and $d|n$.
- (3) $k|m$ and $k|n \Rightarrow k|d$.

We say that m and n are **coprime** or **relatively prime** if $\gcd(m, n) = 1$.

Examples $\gcd(36, 84) = 12$

$$\gcd(-9, 15) = 3$$

$$\gcd(9, 16) = 1 \rightarrow 9 \text{ and } 16 \text{ are coprime integers.}$$